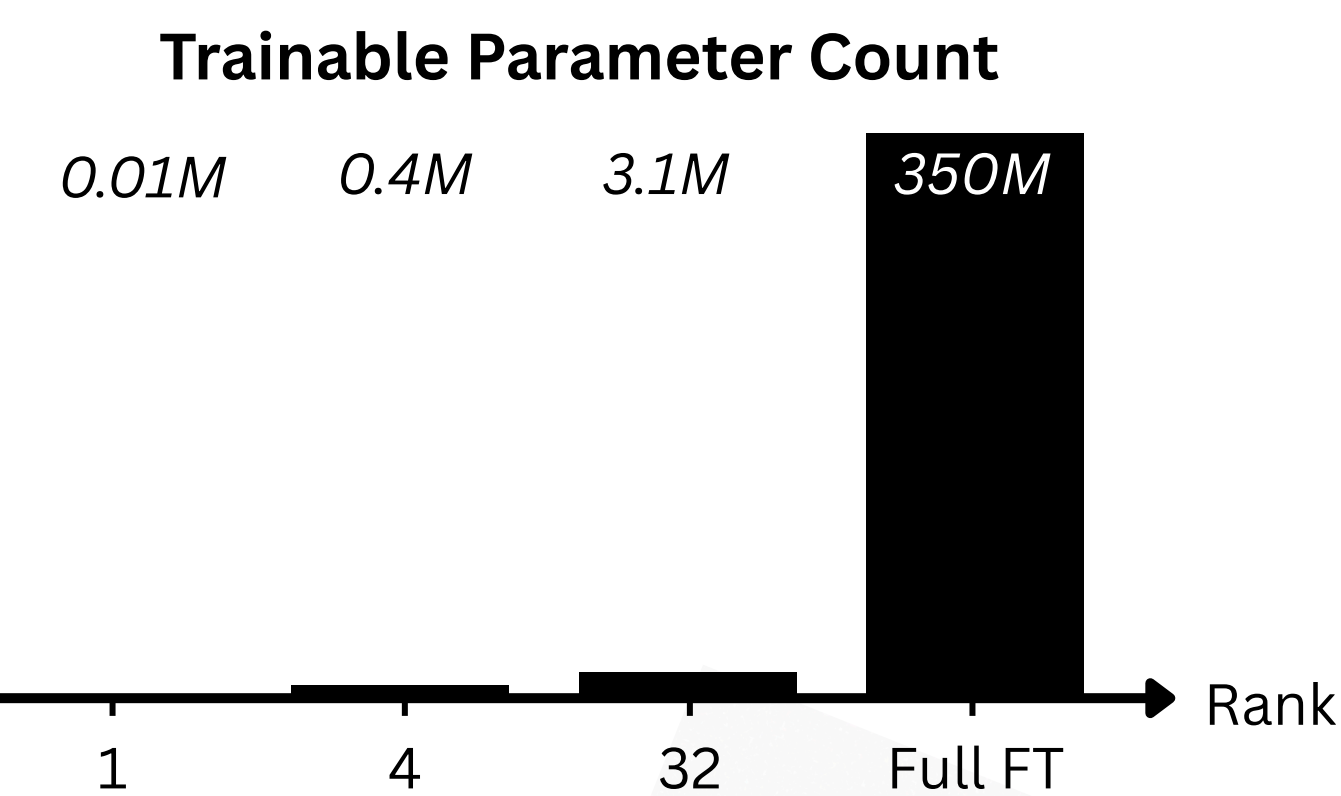


LoRA: Low-Rank Adaptation

of Large Language Models

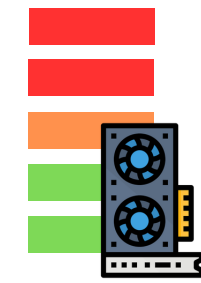
Hu et al., 2021



01 Introduction

Full finetuning is **EXPENSIVE!**

- How to reduce # of trained parameters without reducing performance?
- Previous methods reduce context length or add inference latency



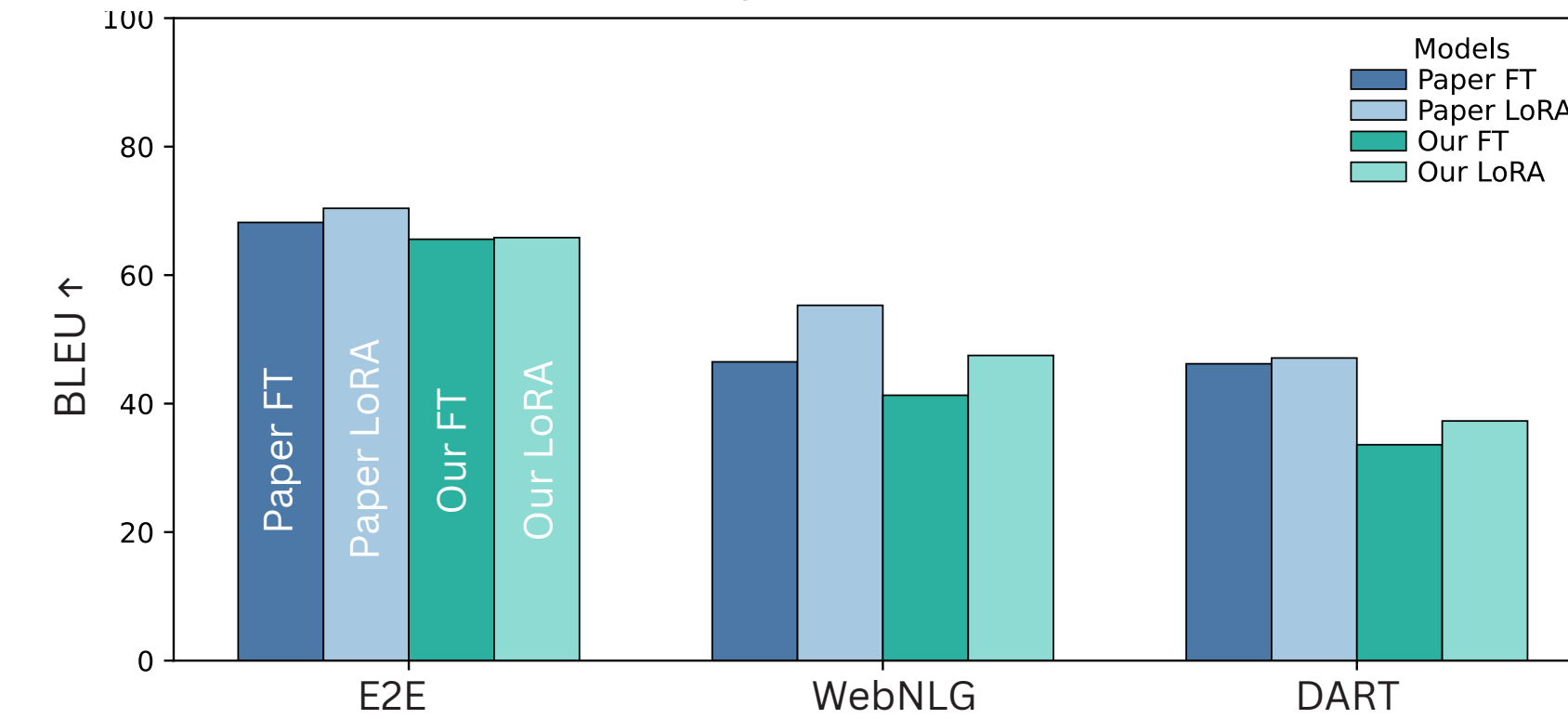
03 Experiments

Train a full finetune (FT) baseline and LoRA on each dataset

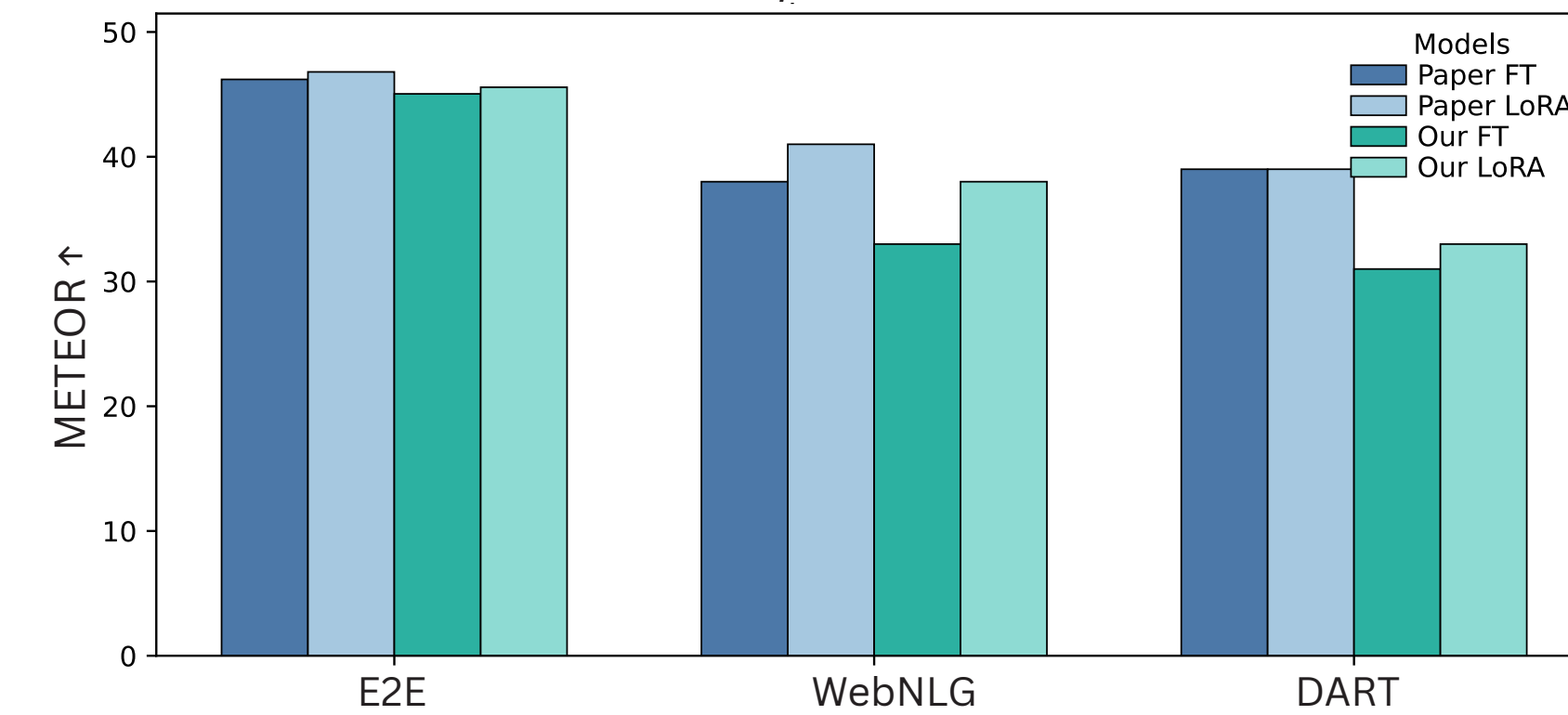
- trainable parameter count
- peak GPU memory
- accuracy, loss, throughput
- Rank ablation for best LoRA rank $\{= 1, 2, 4, 8, 16, 32\}$

04 Results

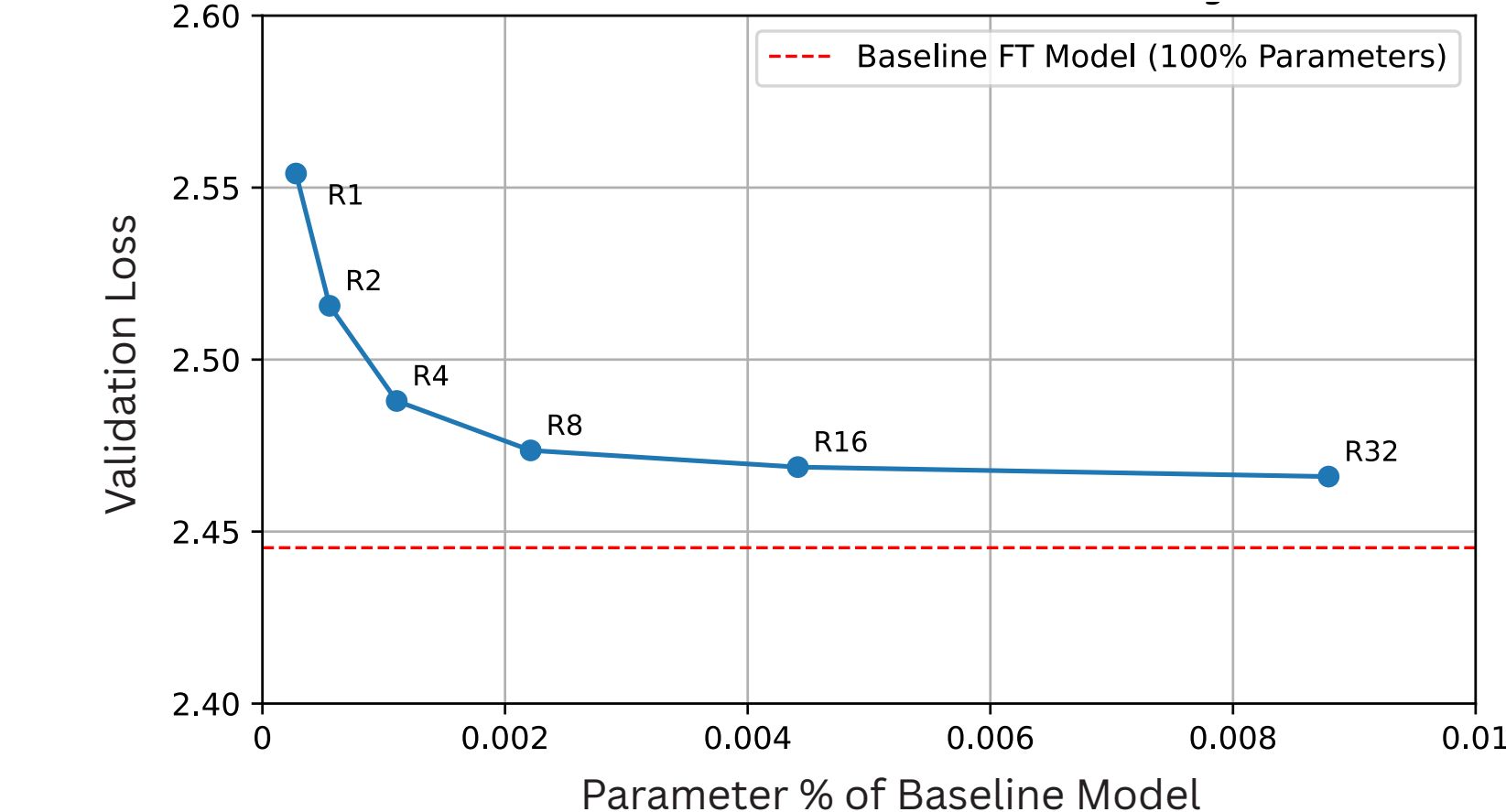
BLEU Comparison Across Datasets



METEOR Comparison Across Datasets



Validation Loss vs Parameter %



05 Tables

E2E Dataset

Model	BLEU ↑	NIST ↑	METEOR ↑	ROUGE-L ↑	CIDEr ↑
FT Paper	68.2	8.62	46.2	71.0	2.47
LoRA Paper	70.4	8.85	46.8	71.8	2.53
Our FT	65.5	6.96	45	64.7	1.73
Our LoRA	65.8	6.94	45.6	65.2	1.81

WebNLG Dataset

Model	BLEU ↑			METEOR ↑			TER ↓		
	U	S	A	U	S	A	U	S	A
Paper FT	27.7	64.2	46.5	0.30	0.45	0.38	0.76	0.33	0.53
Paper LoRA	46.7	62.1	55.3	0.38	0.44	0.41	0.46	0.33	0.39
Our FT	14.0	56.1	41.3	0.17	0.43	0.33	0.85	0.45	0.60
Our LoRA	40.3	51.6	47.5	0.36	0.39	0.38	0.57	0.48	0.51

DART Dataset

Model	BLEU ↑	METEOR ↑	TER ↓
Paper FT	46.2	0.39	0.46
Paper LoRA	47.1	0.39	0.46
Our FT	33.6	0.31	0.68
Our LoRA	37.3	0.33	0.63

902x fewer parameters!

1.4x less gpu memory!

02 Methodology

Datasets	Model
E2E NLG Challenge WebNLG DART	GPT2-Medium (355M)

General Dataset Example:

Structured Meaning Representation

name[Blue Spice], eatType[coffee shop], area[city centre]

delim

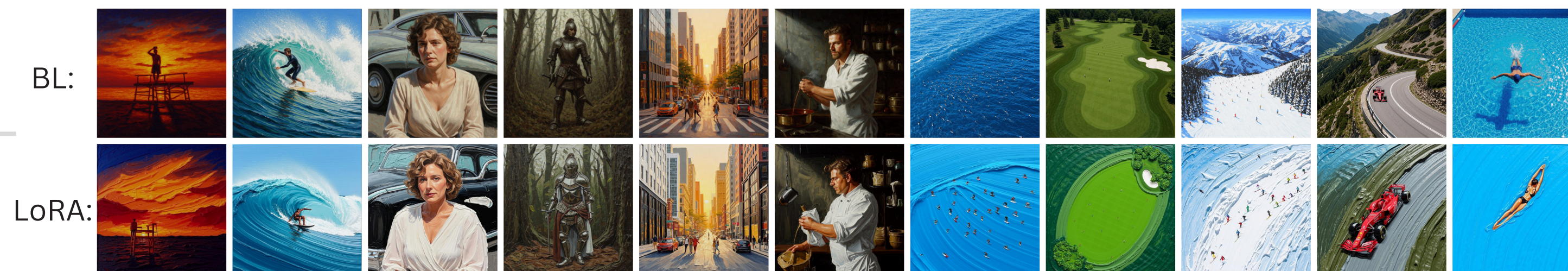
<|SEP|>

Reference

A coffee shop in the city centre area called Blue Spice.

06 Diffusion LoRA

LoRA on Flux.2 Klein 9B – Baseline vs. Lora



07 Conclusion

- LoRA able to achieve similar or better performance than full finetuning on generation tasks
- At a **significantly reduced training cost!**

08 References

Hu, Edward J., et al. LoRA: Low-Rank Adaptation of Large Language Models. 2021. arXiv:2106.09685.

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